

Updated Technical Work Plan for IROS Resubmission

*Distributed Control Framework for Connectivity Preservation and Collective Reconfiguration
in Flow-Driven Robot Fleets*

Prepared: Jan 27, 2026 (MST, America/Phoenix)

Due: Jan 28, 2026 12:00 PM (MST)

Planning horizon: Jan 27, 2026 → Feb 6, 2026

1. Objective and constraints

Objective: Produce an IROS-ready resubmission with rigorous model/problem statement, explicit theoretical guarantees (proofs), and supporting simulation artifacts that directly address the prior ACC reviews.

Work priorities (locked for this window): (1) targeted literature review → (2) proof-driven math + theorem package → (3) implementation + simulation suite → (4) paper editing.

2. Reviewer → technical roadmap (issue → theorem/proof → simulation artifact)

1. R1-1 Model confusion (Sec. II-A vs II-B)

- Fix: Unify to a single control-affine dynamics model used for analysis, and explicitly label any higher-fidelity simulation model as a specialization. Clarify state, inputs, disturbances, and what each robot measures/knows.
- Theorem link: No new theorem required, but all later proofs will reference the unified model.
- Simulation/empirical artifact: Artifact: updated Sec. II with a single consistent model + assumptions block; sanity-check simulation uses the same model.

2. R1-2 Flow term identical across robots → connectivity 'trivial'

- Fix: Replace the centroid-flow term with spatially varying flow $f_{\text{flow}}(x_i, t)$ (or explicitly model flow-gradient effects as bounded relative disturbance). Add assumptions (bounded/Lipschitz flow, bounded disturbances) and show the relative dynamics are non-trivial.
- Theorem/guarantee link: Proof: robust CBF invariance under drift + disturbance (see Theorem T1).

- Simulation/empirical artifact: Artifact: new simulation case with spatially varying flow (vortices/gradients) where uncontrolled team loses connectivity; proposed method preserves it; baseline comparison to existing pairwise-range CBF.

3. **R1-3 Problem statement not rigorous**

- Fix: Rewrite problem statement in math: define graph $G(t)$, neighbor sets, safe set (collision + connectivity constraints), objective set (goal/target), and what constitutes success.
- Theorem link: Proof: main correctness theorem references these definitions (Theorem T3).
- Simulation/empirical artifact: Artifact: revised Problem Statement with formal definitions and notation; include a compact list of assumptions A1–A6.

4. **R1-4 No proof of correctness**

- Fix: Add a guarantees section with explicit theorem statements + proofs/sketches: (i) pruning preserves connectivity, (ii) CLF–CBF–QP preserves safety/connectivity, (iii) goal convergence under CLF (with relaxation).
- Theorem/guarantee link: Proof: Theorems T1–T4 (outlined below).
- Simulation/empirical artifact: Artifact: 2–3 theorems + concise proofs in paper; detailed derivations kept minimal to meet page limits.

5. **R2-1 Provide convergence/stability guarantees**

- Fix: Formalize CLF inequality used in the QP and prove convergence (or practical convergence under relaxation).
- Theorem/guarantee link: Proof: Theorem T2 (goal convergence/practical stability).
- Simulation/empirical artifact: Artifact: goal-reaching plots + convergence curves; optional ablation (with/without CLF).

6. **R2-2 Define $d_{\min}$**

- Fix: Define $d_{\min}$ as the minimum allowable inter-robot separation for collision avoidance; connect to h_{safe} .
- Theorem/guarantee link: No new theorem beyond safety invariance.
- Simulation/empirical artifact: Artifact: add definition in Sec. III-B / Simulation parameters; show no collisions in results.

7. **R2-3 Where is Lyapunov used?**

- Fix: Show the CLF constraint explicitly in the QP and derive $\dot{V}$ condition; explain when it is active/inactive.
- Theorem/guarantee link: Proof: included in Theorem T2.
- Simulation/empirical artifact: Artifact: explicit QP formulation in paper; include $V(t)$ plots.

8. R2-4 Add more relevant references

- Fix: Expand introduction/related work: CBF connectivity, tree/MST-based constraint reduction, distributed edge-criticality/bridge detection, connectivity under disturbances/time delays.
- Theorem/guarantee link: No theorem.
- Simulation/empirical artifact: Artifact: updated related-work paragraph + 8–12 added citations.

3. Theorem/proof package to be produced (minimum set for acceptance)

The goal is to keep proofs compact and aligned with standard CBF/CLF results while making the new pieces explicit (flow-gradient handling + pruning correctness).

9. T1 — Robust safety + connectivity forward invariance (CBF)

- Show that for each active safety constraint $h_{\text{safe_ij}}(x) = ||x_i - x_j||^2 - d_{\text{min}}^2$ and connectivity constraint $h_{\text{conn_ij}}(x) = R_{\text{max}}^2 - ||x_i - x_j||^2$, the chosen QP constraints render the superlevel set forward invariant under $\dot{x}_i = f_{\text{flow}}(x_i, t) + u_i + d_i$, with bounded/Lipschitz flow and bounded disturbances. Result: no collisions and all critical edges remain within range.

10. T2 — Goal convergence / practical stability (CLF)

- For target-assigned robot(s), define $V_i = ||x_i - x_{\text{goal}}||^2$. With a (possibly relaxed) CLF constraint in the QP, prove exponential convergence when feasible; otherwise practical convergence with bound determined by relaxation penalty.

11. T3 — Global connectivity from maintaining a connected critical backbone

- If the maintained edge set $E'(t)$ is connected (e.g., a spanning tree/backbone) and all edges in E' remain within range via CBFs, then the full communication graph $G(t)$ remains connected.

12. T4 — Pruning algorithm correctness (connectivity preserved during edge removals)

- Prove that the pruning stage never disconnects the network: (i) critical edges are never removed; (ii) any removed non-critical edge has a certified alternative path and batching avoids simultaneous vertex-adjacent deletions. Conclude G' remains connected and edges monotonically reduce.

4. Simulation artifacts (reproducible evidence package)

- Code branch tagged with the IROS-resubmission version; scripts to reproduce all figures.

- Scenario S1 (Connectivity stress): spatially varying flow field + disturbances; compare uncontrolled vs proposed controller; metric: connected-ratio over time, min link distance, collision events.
- Scenario S2 (Constraint-reduction benefit): full pairwise CBF vs tree/backbone CBF; metric: control effort ($\int ||u||^2$), QP constraints count, runtime per step, connectivity success.
- Scenario S3 (Target engagement): emerging target detection; measure time-to-target, team displacement, connectivity maintained; compare to centralized MST backbone baseline.
- Optional sweep: $N \in \{6, 12, 20\}$, flow strength sweep; report success probability and control cost scaling.

5. Day-by-day milestones and deliverables (Jan 27 → Feb 6)

Deliverables are concrete artifacts (notes, theorem drafts, code commits, figures, PDFs). Daily work is planned within office hours; evening work is limited to light review/organization.

Date	Technical focus (what to do)	Concrete deliverables by end of day
Tue Jan 27	• Lock technical roadmap: decide final dynamics model + assumptions; outline theorem package (T1–T4). • Targeted literature skim: CBF invariance/robust CBF; CLF–CBF QP feasibility; distributed critical-edge detection; MST/tree-based constraint reduction. • Draft the written plan document (this plan) to ~80% complete.	• Reviewer→Fix→Theorem→Sim matrix completed (1 page). • Theorem list with assumptions A1–A6 + proof outlines (1–2 pages). • Plan draft v0.8 (ready for final polish on Jan 28 morning).
Wed Jan 28	• Finalize and submit the written plan by 12:00 PM. • After submission: write precise math problem statement + notation; begin drafting T1/T3 with correct derivatives under flow.	• Submitted plan (PDF/doc). • Rewritten Problem Statement • Draft theorem statements T1–T4.

Thu Jan 29

- Deep literature reading for proofs: robust CBF under disturbances; QP feasibility/continuity; graph connectivity results used (spanning tree implies connectivity).
- Write proofs: complete T3/T4 first (graph-theoretic), then T1 (CBF invariance).
- T3 + T4 proofs drafted (complete).
- T1 proof skeleton with all required bounds (flow Lipschitz/disturbance terms identified).
- Reference shortlist (10–15 papers) with 1–2 line summaries.

Fri Jan 30

- Finish T1 (robust invariance) and T2 (CLF convergence/practical stability) proofs.
- Integrate proofs into a compact Guarantees section (space-aware).
- Finalize simulation plan + figure storyboard (what plots, what metrics, baselines).
- Guarantees section v1 complete (T1–T4 in LaTeX).
- Simulation experiment checklist + figure storyboard.
- Updated bibliography entries ready to paste.

Sat Jan 31

- Light but focused writing: tighten assumptions; reduce proof length; produce clean notation table.
- Draft related-work paragraph expansion + new citations list.
- Polished theorem statements + compact proofs (v2).
- Related-work expansion draft + citation list.

Sun Feb 1

- Simulation readiness: enumerate parameters, scenarios, and expected outcomes; pre-build config files; plan baseline implementations.
- Write an ablation plan (what to compare and why).
- Simulation config templates + run list (ready to execute Monday).
- Ablation plan (1 page).

Mon Feb 2	• Implementation day 1: update dynamics to spatially varying flow $f_{\text{flow}}(x_i, t)$; implement robust CBF inequality terms; verify QP feasibility on small cases. • Run quick sanity simulations (small N) to validate invariance and no-collision.	• Code commit: updated dynamics + robust CBF constraints. • Sanity run logs + 1–2 debug plots confirming constraints hold.
Tue Feb 3	• Simulation day 2: run S1–S3 full experiments; tune gains; log metrics; generate preliminary figures. • Start drafting results text (bullet form) tied to each reviewer concern.	• All core simulations completed (or queued with saved seeds). • Figures v1 (PDF/PNG) + metrics table v1. • Results bullet draft linked to reviewer items.
Wed Feb 4	• Writing day 1: integrate revised model/problem statement, proofs, and new results into the paper. • Create a ‘response-to-review’ checklist to ensure every comment is explicitly addressed in text/figures.	• Full paper draft v1 (complete end-to-end; may be rough). • Reviewer checklist marked with section/figure references.
Thu Feb 5	• Writing day 2: improve clarity; tighten proofs; ensure Lyapunov/CLF usage is explicit; define $d_{\min}$ and all parameters. • Formatting: enforce IROS/IEEE template, page limit, and figure readability.	• Near-final paper PDF v0.9 (page-limit compliant). • Final figures and captions. • Final reference list.
Fri Feb 6 (deadline)	• Final validation: rerun one representative simulation per scenario to confirm reproducibility; finalize plots if needed. • Submission packaging:	• Submission-ready paper PDF + reproducibility package snapshot (tagged). • Short internal changelog + run instructions.

final proofread, PDF checks,
video link verification (if
used).